Freight rate prediction

Chronological validation, robust modeling and a reproducible submission

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October MAE	October MAPE	Baseline MAE	MAE reduction
$100.26	**4.70%**	$231.30	**56.7%**

Evaluation design

The final inputs cover November-December 2025, after the January-October labeled data. I therefore validated forward in time, keeping every date in one partition. Model selection used pooled dollar MAE across two future months.

Purpose	Fit period / rows	Evaluate or predict / rows
Choose model	Jan-Jul / 33,718	Aug-Sep / 9,429
Locked evaluation	Jan-Sep / 43,147	October / 4,853
Two-month stress check	Jan-Aug / 38,477	October / 4,853
Final fit	Jan-Oct / 48,000	Nov-Dec / 12,000; fixed chart / 31

Candidate comparison

January-July fit; August-September evaluation	MAE ($)	RMSE ($)
Equipment x distance baseline	235.21	663.05
Quote signal x distance	195.66	663.31
Huber additive, one annual harmonic	109.45	616.87
Huber additive, two annual harmonics	167.17	642.08
CatBoost, core inputs (selected)	90.25	612.87
CatBoost + market index	97.90	615.88
CatBoost + market index + quote	134.80	621.08

All CatBoost ablations share fixed settings. The winner then evaluated October without retuning: RMSE $645.76, R2 0.822. The day-block bootstrap 95% MAE interval is $84.69-$116.36 (2,000 resamples; conditional on this month). Initial descriptive EDA summarized the full labeled file, so October was held out from model selection and fitting rather than fully blinded.

Error behavior

October median absolute error is $26.67 and its 95th percentile is $141.66. The largest 1% of absolute residuals account for 97.4% of squared error. No evaluation rows were removed. The fixed two-month horizon check gives $99.12 MAE and 4.65% MAPE; it did not change the model.

Data quality and model

Train-only cleaning and a stable input contract

Issue	Labeled data	Final prediction data	Treatment
Missing weight	300	165	Training-fold equipment median; global fallback
Negative weight	292	145	Treat as invalid; same imputation; add flag
Missing market index	374	249	Exclude from selected model
Duplicate IDs / rows	0 / 0	0 / 0	Fail on duplicate prediction IDs
Long target tail	Max $25,533	No labels supplied	Retain all labeled rows; robust MAE loss

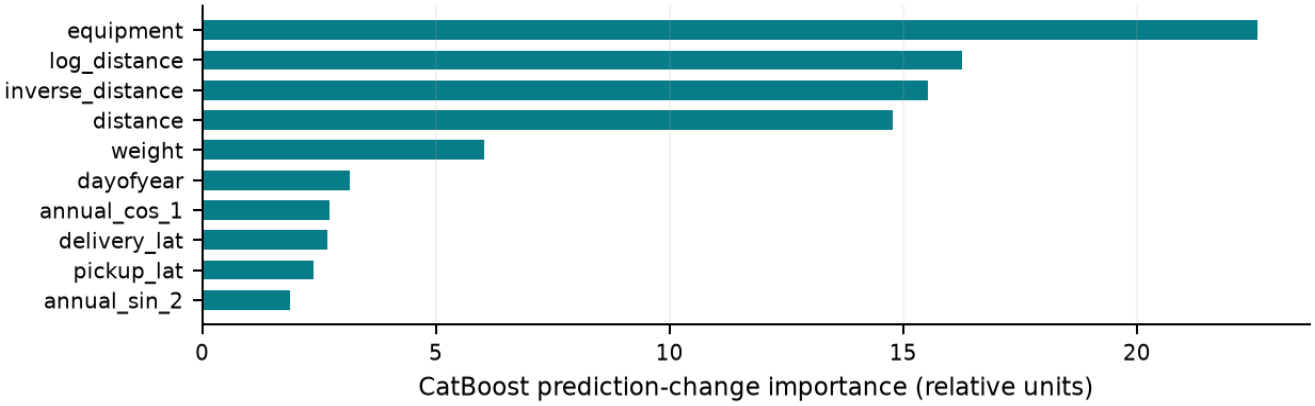
Why exclude the auxiliary signals?

The quote-derived baseline has $317.26 MAE in August but $71.75 in September. That changing reliability suggests a proxy or availability risk; it does not prove leakage. The matched CatBoost ablations confirm that including the signals hurts the two-month development result. Both are also absent from the fixed December table. The final model uses the same six-input contract for every prediction.

Estimator and feature engineering

CatBoost predicts rate per mile with MAE loss and distance-proportional weights. For positive distance, weighted absolute error in rate per mile equals dollar absolute error up to a fixed scale. The output is multiplied by distance and floored at $1. Settings: 1,600 iterations, depth 6, learning rate 0.045, L2 8, seed 42, four CPU threads.

Features include directional city/lane categories, equipment, cleaned weight and its flag, distance/log/inverse distance, weekday/weekend, day, month, day of year and two annual sine/cosine harmonics. City coordinates are derived from fitting-fold lookups. No load ID, manually computed target aggregate, validation label or auxiliary signal enters the feature table. CatBoost learns its categorical statistics inside each training fit. The tree count is fixed; October is not used for early stopping.

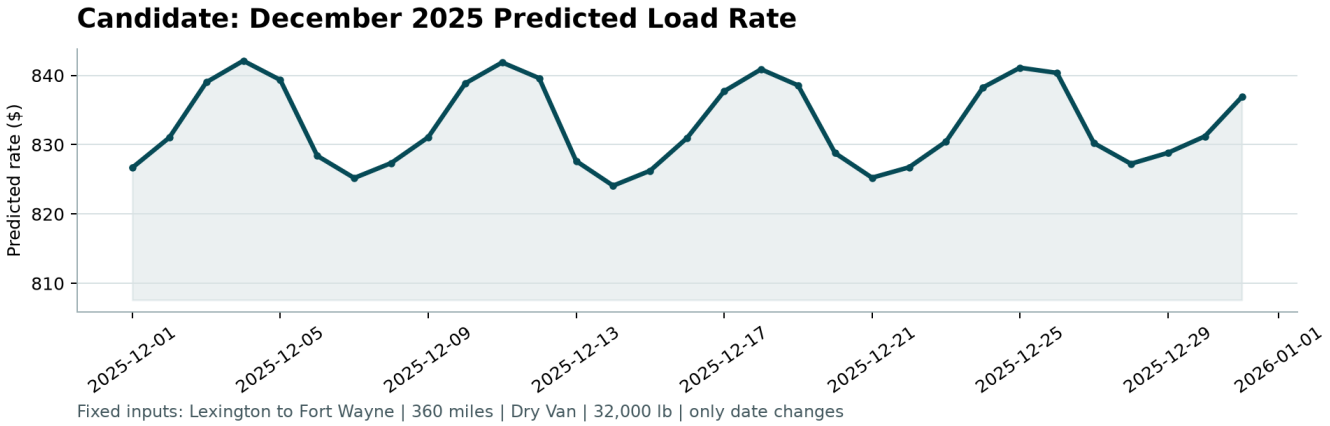


Final-refit feature importance is descriptive, not causal and not a separate test. Correlated calendar/location features can share importance.

Fixed December forecast

Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lb

This is the unmodified image produced by the supplied **score.py**. Only the date varies across the 31 rows. The same final model, fitted on all 48,000 labeled loads, produced this curve and the 12,000 load-level predictions.



Daily predicted rates range from $824.08 to $842.10. The scorer validates both CSVs and generates this chart; Spotter calculates hidden-label accuracy after submission.

Reproduction and output integrity

The repository includes source, input setup instructions, the original scorer, requirements and an exact environment lock. Raw assessment datasets stay local. **experiment.py** recreates selection; **train.py** evaluates the locked model, saves diagnostics and refits; **predict.py** matches predictions to template IDs. It preserves December input columns and validates one-to-one ID coverage and finite positive rates. Both CSV outputs are written to cents. Four contract tests and the original scorer pass.

Run after installing requirements.txt
python train.py
python score.py --predictions validation_predictions.csv --december-predictions data/december_chart_inputs.csv
python make_report.py
python -m unittest discover -s tests -v

Limits and next steps

Only ten labeled months from one year are available. Seasonal features cannot establish December holiday peaks or future market shocks. The chart has no observed December prices for this fixed lane and should not be interpreted as validated daily seasonality. Repeat cities and lanes are allowed across time splits, matching the forecasting use case. About 12.2% of final directional lanes were unseen in training; unseen-city or new-year performance was not validated. Monitor future dollar error and drift once new labels arrive.

Submission: repository, validation_predictions.csv, this report, and a candidate-recorded 2-3 minute camera-on Loom. A timed walkthrough guide is included; the actual recording is a remaining candidate action.